Class 15 — Discussion Notes

1 Two social aggregation questions

1. How do we select leaders, or constitutions, or policies, given votes?
2. How well is a society doing given how well its members are doing?

These are related mathematically, and it's often easiest conceptually to think about the first, but we really care about the second.

- Sen's list of what the field covers runs well past voting: when majority rule gives consistent answers, how to accommodate rights and liberties, how to measure poverty across people whose predicaments differ, how to value a public good like the environment or epidemiological security.
- This has shaped a bunch of Sen's influential work on other fields: famines, gender inequality, and a theory of justice.
- Aristotle and Kautilya, both fourth century BC, both wrote on how a society decides. Useful against "this is a modern Western preoccupation".

2 Condorcet's paradox

New Introduction, Early Origins (p. 3).

- Person 1: x, y, z. Person 2: y, z, x. Person 3: z, x, y.
- x beats y, y beats z, z beats x, each two votes to one.
- No winner, and the social ranking generated by majority rule.

Condorcet was one of a number of people working on social choice in the 1780s, then one of many, many people working on political reform in the Revolution. He voted against executing the King, but he did think that the King should be a galley slave, so not exactly a moderate. He did not survive the Revolution.

3 Positivism, and the shrinking informational base

New Introduction, Crisis in Welfare Economics.

- Traditional welfare economics was utilitarian: Edgeworth, Marshall, Pigou, from Bentham (and both Mills, though Sen doesn't focus on them as much) rather than from Borda and Condorcet.
- Utilitarianism is already distribution-blind. Sen's example: someone with a lower capacity to convert income into utility gets a *smaller* share in the ideal utilitarian world.
- Then Robbins, explicitly influenced by logical positivism removes interpersonal comparison has no scientific basis. **"Every mind is inscrutable to every other mind and no common denominator of feeling is possible"** (Robbins 1938, p. 636; quoted around p. 10). (How are individual minds scrutable? By watching them make decisions. If you choose Coke over Pepsi,

I know you prefer Coke to Pepsi; that's the kind of intra-personal utility comparison that is allowed.)
- Result: the base shrinks from sum-totals to bare rankings, which is Borda and Condorcet's voting space. Bentham and Robbins make distinct shrinks, but the effect compounds.

4 Arrow's theorem

- **Where the question came from.** 1948, Olaf Helmer at RAND, worried about applying game theory to international relations because "the players were countries, not individuals". So he asks a graduate student: in what sense could collectivities be said to have utility functions? Arrow answers by pointing at Bergson's 1938 social welfare function, sets out to write the explanation, and becomes convinced it cannot be done. The theorem takes "about three weeks" (p. 272). He abandons an advanced dissertation in mathematical statistics; the short version is "A Difficulty in the Concept of Social Welfare", 1950.
- **The four axioms, roughly stated.** Universal Domain: works for every profile. Pareto: if everyone prefers x, society does. Non-dictatorship. Independence: ranking x against y uses only preferences between x and y.

5 5. Black-boxing, and why Arrow's Theorem and Condorcet's Paradox are different

This is mostly from A1, and the technical details are in A1* which we'll get to next time.

- Nothing in the axioms says we may not look at what the social states are actually *like*, whether they are equal or unequal, whether they respect rights and liberties or not, etc.
- But the principles other than Dictatorship prove **Spread of Decisiveness** (p. 274, proved in A1*): if a group is decisive over one pair, it is decisive over every pair.
- Universal Domain, Independence and Pareto together therefore turn the social states into undescribed boxes.
- So it makes no difference how much inequality one state contains rather than another, or how much poverty, or whose rights are violated.
- Arrow's theorem is a result about an informational restriction. It is not a result about cycles in voting.

6 Why Welfare economics and voting might be different should not have wanted voting rules anyway

A1, *Welfare Economics and Political Rules* (p. 275-6).

- Sen's second reason for resisting the Condorcet reading.
- Even if majority rule were perfectly transitive, it would be a poor instrument for welfare economics. Transitivity failure is not its worst problem there.
- **"Was Arrow taking too big a leap by answering Helmer's political question with Bergson's welfare-economic answer?"**

- This is why I started with the 2 questions. A good voting rule might be a bad welfare rule. A policy that makes 60% a little better off and 40% much worse off might be supported by a majority if they are sufficiently uncaring. But it would not increase social welfare.
- That last claim is a bold assertion - *do you agree*?

7 Incomplete social orderings

- Drop completeness and the impossibility weakens, but not by much.
- Suzumura consistency: allowing a little incompleteness in the social ranking also disestablishes the result. Sen's gloss is that this shows how finely cut Arrow's consistency demand has to be.
- Needed first: **indifference is not incompleteness.** Take "x is at least as good as y" and "y is at least as good as x". Indifference asserts both; incompleteness asserts neither. Ordinary language runs them together. We might discuss the small improvement argument here.

8 Dropping transitivity, and cardinal or interpersonal comparison

- **Quasi-transitivity** (strict preference transitive, indifference not) invalidates Arrow. Sen notes he reported this "more as a curiosity" than as a resolution.
- Gibbard showed immediately what you get instead: an **oligarchy**, a group jointly decisive with a veto for each member. (The Gibbard here is Allan Gibbard, long time member of Michigan philosophy. He has two particularly famous results in economics; this is one of them.)
- A decade of papers followed in which each weakening produced a new infelicity, and each infelicity was ruled out by a new axiom, producing a new impossibility.
- **Cardinality and interpersonal comparison** are the other way to widen the base, and they solve the problem at the cost of taking on everything Robbins thought was unscientific.

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9 If we have time

Three things "preference" might mean. Votes and ballots (Borda, Condorcet); interests (Hume, Bentham, Mill); judgements and moral sentiments about society (Smith, Kant). Sen uses the one word, as the field does, and warns against letting the word make them the same thing. Connects directly to the initial point about the two questions.

Sen's anti-holism, and how fast he moves. Ch 1 opens by rejecting the idea that society is an entity with preferences of its own: "Anyone who finds his fulfilment in this assumption is entirely welcome to it, and this book must bore him." He has Marx on his side: "What is to be avoided above all is the re-establishing of 'Society' as an abstraction vis-a-vis the individual." But there are many others, notably following Rousseau, who will not like this.

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10 Notable quotes to have to hand

> There is something in common between singing romantic songs about an abstract motherland and doing optimization exercises with an arbitrary objective function for a society. While both the activities are worthy, and certainly both are frequently performed, this book, I fear, will not be concerned with either.

On the range of the subject: the calm economic technician who says that taxing commodity a would be inoptimal, and the crowd outside the Bastille on 14 July 1789 shouting "Down with the second drawbridge!", are both making judgments about collective choice.

On reflexivity: so mild that it is "best looked at as a condition, I imagine, of sanity rather than of rationality."

On completeness: desirable, but "we shall not make a fetish of it."

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11 Where this is going

Sen's **four departures** from Arrow's framework, set out in Ch 1 §1.2, are the architecture of the book.

1. The society need not have an ordering (Ch 4)
2. It need not be complete (Ch 2, 8, 9)
3. It might rely on intensities and cardinality (Ch 7, 8)
4. It might rely on interpersonal comparison (Ch 7, 9)

Next time: the apparatus, then unanimity.